Amazon River: Solution for Hunger and Drought

João Carlos Holland de Barcellos

Abstract: This essay aims to demonstrate the economic feasibility of “transposing” part of the waters of the Amazon River to irrigate the semi-arid regions of northeastern Brazil, with the objective of making the region fertile for agriculture. The waters of the Amazon River [1] would be captured at its mouth, just before entering the Atlantic Ocean, and pumped by one or more nuclear power plant(s), specially built for this purpose, to the northeastern semi-arid region.

Keywords: *Drought, Northeast, Hunger, Arid, Transposition, Amazon River, Nuclear Energy, Agriculture, Water.*

The Value of Waste

To get an idea of the cost of fresh water, we should obtain data on the cost of desalination (the process that removes salt from seawater):

“Currently there are 7,500 plants in operation in the Persian Gulf, Spain, Malta, Australia, and the Caribbean, converting 4.8 billion cubic meters of saltwater into fresh water per year. The cost, still high, is around ***US$ 2.00 per cubic meter****. The large plants, similar to oil refineries, are located in Kuwait, Curaçao, Aruba, Guernsey, and Gibraltar, supplying them entirely with fresh water extracted from the sea.” [2]*

However, there are technologies that promise to do the same for around US$ 0.50 per cubic meter:

“The cheapest desalinated seawater is found in Israel, where the world’s largest reverse osmosis plant was built on the Mediterranean coast, in Ashkelon. It produces 270,000 cubic meters of water per day. Israel’s policy on water is notoriously opaque, but the government guarantees it can supply water at around US$ 0.50 per cubic meter. This represents about one-third of the production cost in Saudi Arabia, and one-sixth of the typical desalination cost in effect 20 years ago.” [3]

If we use this lower desalination value (US$ 0.50/m^3) to evaluate the fresh water from the Amazon River that flows into the ocean, we can find the value that is being wasted:

The flow at the mouth of the Amazon River is 216,342 m^3 per second [1]. This means approximately 18 billion cubic meters of fresh water are being salinated per day. This volume of water, if desalinated, would have a cost of at least **US$ 9 billion per day**!

Thus, we can appreciate the enormous daily waste of water resources that could be used before flowing into the sea.

My idea is to estimate the technical and financial feasibility of the project, quantifying and comparing the investment required to transpose this water — just before it enters the sea, without harming those who already use the river’s waters or the ecosystem it sustains along its course — against the gains obtained if the water were used for agricultural irrigation in the northeastern semi-arid region.

Although this idea occurred to me in January 2000 [18], Mangabeira Unger, in 2008, also raised the same banner: “In one region, water is wasted uselessly. In the other, it is catastrophically lacking.” When asked after the first of a series of meetings about his major projects, he commented: “Brazil needs to stop being afraid of ideas.” [23]

Water Demand in the Semi-Arid Region

Let us now determine the amount of water needed to irrigate the entire northeastern semi-arid region.

The total area of the Northeast is 1,558,196 km², and the semi-arid region corresponds to about 80% of that total [4].

Therefore, the area of the northeastern semi-arid region is approximately: 1,200,000 km².

We will base our calculation of the necessary water demand on a study conducted in the Paracatu river valley region:

“The average amount of water needed to meet the demand of irrigated crops in the Paracatu basin was ***0.35 L s⁻¹ ha⁻¹****, with the value corresponding to the month of highest evapotranspirometric demand being 0.50 L s⁻¹ ha⁻¹.” [5]*

Converting this value to m³/km²/s we obtain: 0.035 m³/s/km².

With this value we can now estimate the total demand for the entire semi-arid region:

Total Demand = Semi-arid area × Demand per km² = 1,200,000 × 0.035 = **42,000 m³/s** (= 1.3 trillion m³/year).

That is, we will need approximately 42,000 cubic meters of fresh water per second to irrigate the entire region. We can also note that the São Francisco River would not be adequate for this task, since its flow at the mouth is 1,850 m³/s [6], meaning approximately twenty-two “São Francisco Rivers” would be needed for such an undertaking. The Amazon’s waters could supply approximately four times this area. Moreover, it is nearly impossible, politically speaking, to take water from those who already use it to give to those who need it most — as demonstrated by the São Francisco River transposition, which faces enormous political resistance. The Amazon estuary waters that would otherwise be salinated will not be claimed by anyone, since they cannot be used, and therefore there would be much less political resistance.

Potential Agricultural Value of the Region

Let us now estimate the market value if the entire semi-arid region (1,200,000 km²) were used for agriculture. For this estimate, let us consider soybean, rice, and corn crops, which have grains with high durability and easy storage. The maturation time, from planting to harvest, for all three grains is approximately 120 days, so up to three harvests per year are possible.

For these grains we have the following productivity table [13] [14]:

Rice: approximately 7.0 tonnes per hectare

Corn: approximately 4.81 tonnes per hectare

Soybeans: approximately 2.79 tonnes per hectare

If we divide the total area equally among the three crops (400,000 km² each), we will have 40,000,000 hectares for each grain, giving us a total of:

280 million tonnes of rice, 192 million tonnes of corn, 111 million tonnes of soybeans per harvest. Considering that we can have three harvests per year, and using international market prices (US$ 600 rice [15], US$ 165 corn [16], US$ 400 soybeans [17]), we will have:

840 million tonnes of rice per year, equivalent to US$ 504 billion;

576 million tonnes of corn per year, equivalent to US$ 96 billion;

333 million tonnes of soybeans per year, equivalent to US$ 133 billion.

For comparison:

"In 2005 the world population was already 6.453 billion, world grain production reached 2,219.4 billion tonnes over a harvested area of 681.7 million hectares, per capita production was 0.344 tonnes, and the per capita harvested area was 0.106 hectares." [19]

This would give a total potential gross value of 1.7 billion tonnes per year, at a value of **US$ 733 billion per year**. Equivalent to about 34% of Brazil's GDP and about 10 times Brazil's total grain production in the 2008 harvest of 144 million tonnes [20].

We can also calculate profit per hectare. From an internet search:

"… Farmer Américo Amano, from Londrina, Paraná, bet on both crops. Doing the calculations, he discovered that corn would be much more profitable this year. The farmer's calculations: for soybeans, the expected yield is 50 bags per hectare. To cover costs, Mr. Américo calculates 30 bags per hectare. The price of soybeans this week in Londrina was R$ 42.00; the profit per hectare will be R$ 840.00. For corn, costs should consume half of the 120 bags per hectare he expects to harvest. The price that week was R$ 22.00 per bag. The profit would then be R$ 1,320.00 per hectare — more than 50% advantage over soybeans." [7]

In January 2008 the exchange rate was 1.8 reais per dollar, giving us:

Soybean profit = US$ 460 / hectare

Corn profit = US$ 730 / hectare

If we consider that the northeastern semi-arid has 1,200,000 km² of area, each hectare is 0.01 km², and take the average of both crops (US$ 835 / 0.01 km²), we will have per harvest:

Average profit per harvest = [US$ 835 / (0.01 km²)] × 1,200,000 = US$ 100 billion

Considering up to three harvests per year:

Average annual profit = US$ 300 billion!

This profit is equivalent to about 40% of gross production value, therefore costs represent about 60% of production.

Since these values are proportional to water flow, we can calculate them per unit of flow (m^3/s) per year:

Total grains per year per m^3/s = (1.7×10^9 / 42×10^3) = 40,000 tonnes/m^3/s/year

Gross value per year per m^3/s = (733×10^9 / 42×10^3) = 17 million dollars/m^3/s/year

Net profit per year per m^3/s = (300×10^9 / 42×10^3) = 7 million dollars/m^3/s/year

Pumping Costs

Let us now calculate the cost of pumping the required water (42,000 m^3/s) from the mouth of the Amazon River.

To compute the required plant power, we need to calculate the average height to which the waters must be raised. For this, we will use the average altitude of the Caatinga, which is the main type of terrain in the region and the primary target for irrigation:

"The Caatinga biome is located in the semi-arid region, covering an approximate area of 1,037,517.80 km², spanning 9 northeastern states (Piauí, Maranhão, Ceará, Rio Grande do Norte, Paraíba, Pernambuco, Alagoas, Sergipe, and Bahia), as well as the northern region of the state of Minas Gerais. This region covers 60% of the Northeast, including northern MG (SUDENE area), and 13% of Brazil, with 56% of the northeastern population (including northern MG) and 16% of the Brazilian population. Of the lands covered by the Caatinga, 50% are of sedimentary origin, rich in groundwater. The rivers are mostly intermittent and the volume of water is generally limited, being insufficient for irrigation. ***The altitude of the region ranges from 0–600m.****" [12]*

Taking the average altitude as the mean of the maximum and minimum values, we obtain an **average altitude of 300m**.

With this value we can calculate the power required to raise the water to this average altitude:

Power = Elevation power + Acceleration power (to reach desired velocity)

Elevation Power = Flow rate × weight × Height

Considering that:

Each 1 m^3 of water = 1,000 kg

1 m^3 weighs 1,000 kg × 10 m/s^2 = 10,000 Newtons

Altitude = 300 m

Flow rate = 42,000 m^3/s

With these data: “Elevation Power” = 42,000 × 10,000 × 300 = 126×10^9 Watts = 126 GW

“Acceleration Power” = Flow rate (mass) × ($Velocity^2$) / 2

Taking a transmission velocity of **1 m/s** (3.6 km/h):

“Acceleration Power” = 42,000 × 1,000 × (1^2) / 2 = 0.021 GW

Total required power = 126 GW + 0.021 GW = 126 GW

To calculate the annual cost of maintaining this power, we need the cost per watt-hour:

*“… Another important factor is that the cost of nuclear energy today is cheaper than other sources. Vincent Gilles, of investment bank UBS, said to The Economist that the cost of German plants is 1.5 cents per kilowatt-hour, but that they can sell this energy for three times more when including credits from Europe’s carbon trading program. The cost of gas-powered energy in Germany is 3.1 to 3.8 cents per kW-hour; coal energy can range from 3.8 to 4.4 cents per dollar. In the United States, coal energy costs 2 cents per dollar per kW-hour, natural gas energy costs 5.7 cents, and **nuclear energy costs 1.7 cents**…” [9]*

The annual cost to maintain this power would be:

Annual Cost = (Power × Time of 1 year) × Energy Value

Power = 126,000 MW

Time of 1 year = 8,760 hours

Cost per MW-hour = US$ 17 (= 1,000 × US$ 0.017)

Annual Cost = 126,000 × 8,760 × 17 = US$ 19 billion per year. This cost would be equivalent to 6% of production profits or 3% of annual gross production value.

In addition to nuclear plants being cheaper per generated MW, Brazil has one of the world’s largest uranium reserves:

“Brazil, owner of the sixth largest uranium reserve on the planet, announced last year that it would begin enriching uranium as of May 2004 through INB (Indústrias Nucleares do Brasil). The government also announced that, within a decade, it intended to begin exporting the product. The uranium extracted today in Brazil is sent to Canada to be converted to gas and to Europe (Germany, the Netherlands, or England) to be enriched, then returned to Brazil.” [10]

Furthermore, nuclear waste already has its destination planned:

“Eletronuclear and the CNEN (National Nuclear Energy Commission) presented on August 18, 2008 to President Luiz Inácio Lula da Silva a proposal for the disposal of nuclear waste that will keep the waste safe for the next 500 years. According to Folha Online, there is a ‘high degree of consensus’ around the proposal, which was not detailed. Lula met on that Monday for more than two hours with 11 ministers to discuss Brazil’s nuclear policy. Sources who had access to the meeting said that the ministers

will present a proposal to the president within 60 days for a new nuclear program in the country." [11]

Cost of the Nuclear Plant(s)

The trend is for nuclear plant costs to decrease as technology advances:

"New, safer projects are more expensive than existing alternatives. Proponents of nuclear energy say it is possible to build new plants at a cost of US$ 1,500 per kW of installed capacity, but more realistic estimates put new complexes at US$ 2,000 per kW." [9]

If we use the higher value (US$ 2,000 per kW), we will need a total of **US$ 252 billion** (126 GW × US$ 2,000/kW) to build a plant with the 126 GW (or 93 plants of 1.35 GW each) needed to pump the river water.

This cost would be equivalent to about 84% of the net profit obtained in one year of production, or 34% of gross production in the same period.

We can compare this cost with the Angra 3 plant:

"Eletronuclear, the state company responsible for Brazil's nuclear energy projects, estimates that the cost of Angra 3 will be R$ 7 billion… When Angra 3 is operating with an installed capacity of 1,350 MW, in 2014." [21]

We will need 93 plants of the capacity of Angra 3 to produce 126 GW, which at the expected prices would be equivalent to R$ 651 billion, or approximately US$ 382 billion. Of course, if we had to build 93 plants, the price of each would be much lower than that of a single one. Furthermore, the cost per kW produced is lower for larger plants.

Cost of Transmission Ducts

To estimate the costs of the pipelines needed to carry the Amazon's waters to the northeastern semi-arid, we need to calculate the length of the route the water must travel and the cost per km.

If we use concrete pipes, as per the public tender below, we will have:

"ITEM 3 – supply of 60 linear meters of reinforced concrete pipe of circular cross-section, for rainwater drainage, 1,200 mm, with 1.50 meters in length, according to NBR 8890:2003-ABNT. The unit price budgeted for the acquisition of material for ITEM 3 is R$ 231.00 (06/22/2004)." [22]

The exchange rate at that time was R$ 3.10 per dollar, so the cost of this pipe per meter in dollars would be:

Price per meter = 231.00 / 3.10 / 1.5 = US$ 15/meter

If we pump the water through these pipes at a velocity of 1 m/s, and considering that each pipe has a diameter of 1.2 m, this gives a cross-sectional area ($\pi \times R^2$) of 1.1 m^2.

Thus the flow per pipe is: Flow = Area × Velocity = 1.1 × 1 = 1.1 m^3/s

Therefore, the number of pipes required to provide the necessary flow of 42,000 m^3/s will be 38,000 pipes (= 42,000 / 1.1).

The distance from Belém, near the Amazon mouth, to Petrolina, which we can consider the center of the Northeast, is 1,500 km. Some pipes will end before the center of the Northeast and others after. So we can use Petrolina as the average distance. In this case the transportation cost will be:

Cost of ducts for transport = 38,000 ducts × US$ 15/m × 1,500,000 m = US$ 0.85 billion.

If we consider that the installation cost per meter of pipe equals the price of the pipe:

Total installed ducts = 2 × US$ 0.85 billion = US$ 1.7 billion.

Labor Costs

Taking as a reference the labor cost of the Itaipu Dam, which at peak construction employed 40,000 workers and lasted 9 years (1975 to 1984), and considering an average salary of US$ 300, we can calculate labor costs:

Labor Cost = 40,000 workers × 9 years × 12 months × US$ 300 = US$ 1.3 billion.

Implementation

The project could be implemented gradually, for example, by installing a single small nuclear plant at the river mouth and sending water to the closest and least costly drought-stricken regions. The profit from this first plant could be used to pay for the construction of the second. The profits from the first and second plants would fund the third. And so on: as production generates profits, that money would be used to build new plants that would serve more semi-arid land, which in turn would contribute to ever-increasing production. In this way, the initial investment would not need to be large, and the project itself could be self-expanding.

Environmental Impact

Since only 21% of the Amazon’s waters would be captured at its estuary, where they can no longer be used, there should be no drying effect in this region. Nuclear power plants, unlike coal- or diesel-based plants needed for pumping water, do not emit CO_2 and do not contribute to the greenhouse effect and global warming. The water transported to the semi-arid region will bring ecological benefits to the area, as it will lower average temperatures, creating a low-pressure zone that would favor the migration of warm, moist air masses from the Atlantic Ocean into the region. This could increase average rainfall and make the region less dependent on Amazon water. It is even possible that, with this new rainfall, Amazon waters might no longer be necessary in a somewhat more distant future. We can thus conclude that environmental costs would be positive.

Conclusion

The mega-project of transposing part of the waters (21%) of the Amazon River to irrigate the northeastern semi-arid region would be economically viable, would contribute positively to the region's climate, would lower food prices on the international market, reduce world hunger, and solve the problem of drought and famine in Brazil.

The estimated total cost of the project would be:

Total estimated cost = Cost of plant(s) + Cost of ducts + Labor cost = US$ 252 billion + US$ 1.7 billion + US$ 1.3 billion = **US$ 255 billion.**

Annual net profit = annual net production profit – energy costs for water transport = US$ 300 billion – US$ 19 billion = US$ 281 billion.

Thus the cost of the project would be recovered in less than one year of production (255 billion / 281 billion), a relatively short time considering the scale of the undertaking.

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